

Banking Queue Waiting Time Prediction based on predicted service time using Support Vector Regression

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Abstract—Prediction using different machine learning approaches have been applied in the last few decades in different areas and research fields. Waiting time is an undeniable fact for every queue and it is very important to develop a system that predicts its duration in real life with minimum error. In this paper we applied several machine learning algorithms and among them we chose Support Vector Regression (SVR) in a real-life Banking queue dataset that contains real-life queues of multiple Banks where we predicted waiting time for everyone in the queue. Moreover, we have compared the result of prediction using SVR with different classifications and clustering methods such as K-nearest-neighbor and K means Clustering. We have shown the feasibility of applying SVR in prediction of waiting time in banking queues of developing countries for everyone, which is applicable, and it performs well in queue analysis.

Keywords—Support Vector Regression, Regression, Queue Analysis, Waiting Time Prediction, Queue Management

I. INTRODUCTION

Over the years waiting in a queue has been one of the tedious tasks all over the world. People face queues almost every day in their life. We often observe queuing at Public service counters, Hospitals, Immigration Service, Airport Security Check, transport ticket counters. Moreover, waiting in the queue is an uncertain fact people dealing with everywhere as a result a queued person does not have any idea how much time it will cost to complete their task from his/her position in the queue. This waiting time often depends on the service time on the counter.

Prediction on this area has already become one of the most important tasks for this research. Using different machine learning approaches on predefined structured data prediction on queue waiting time can be accomplished. For supervised learning different classifiers such as Support Vector Machine, K nearest neighbor and Support Vector Machines can be used to classify the previous data that can be found through some time frames. Also, some common regression machine Learning Algorithms such as Support Vector Regression and Linear Regression can be used for continuous response value. For unsupervised learning clustering algorithm such as K-means can be used to the system to learn from the dataset by clustering the similar data to separate groups. Moreover, intensive study on the dataset is the main cornerstone for this research. Without

preprocessed well defined dataset it is very difficult to predict certain facts and obtain high accuracy rates.

In this paper we are going to discuss on a research model that have been used for predicting waiting time on a queue based on the service time using different machine learning approaches. Different machine learning approaches on a predefined test dataset have been used [10], which contains different information about the people standing or waiting in the queue. Different algorithms are implemented for evaluation on the same dataset and found significant results on predicting waiting time on a queue for everyone. A comparison between the implemented machine learning approaches for classification, regression as well as clustering are illustrated in this research which provides a very understanding answer to the selection of a proper. As a result, this paper will address a valid and accurate solution to the problem of prediction of queue time which will further help in better queue management.

II. BACKGROUND STUDIES

Among various machine learning approaches used all thought our research, Support vector machine (SVM) analysis is one of the popular tools for classification and regression, first proposed by Vladimir & Vapnik along with his colleagues in 1992 [1]. When SVM is applied to a regression problem it is just termed as support vector regression [2]. SVM regression is considered a nonparametric technique because it relies on kernel functions. Various fields such as time series prediction, approximation of complex engineering analyses, convex quadratic programming and choices of loss functions SVR has been applied in [3]. Delay in queue systems in banking sector has been proposed in [13] where the predictive model had been compared with many other popular methods. The model had been compared with models crafted from Queuing Theory, Deep Learning, Gradient Boost Machine [16] and Random Forest [15]. In [14], a queue probability tool had been proposed for a single line queue.

Prediction using SVR got several significant successes previously along the last decade. Especially for time related prediction with real life data, the algorithm quite huge achievements. In a Precise traveling time prediction with a route guidance system that can suggest optimal alternate route or warn traffic have been proposed in [2] where

preprocessed historical data set has been used. Three categories of traffic data have been taken: (historical, current and predictive) for the dataset [4]. Several prediction methods such as SVR Prediction Method, Current Travel-Time Prediction Method, Historical Mean Prediction Method have been tested [2]. Many successful applications have proven that artificial neural networks (ANN) can be a very useful tool to predict time-series or forecasting [5] as well. SVM also has promising results in prediction on stock exchange or financial time-series [6]. LIBSVM software was used to observe the performance experiments when both Gaussian radial basis function and polynomial kernel are used as the kernel function of SVM. In comparison of SVM with BP Network and CBR, SVM got the highest prediction accuracy for the training data. For CBR, to retrieve relevant cases the k-nearest neighbor method was used. Hence the results showed the feasibility of SVM in financial time series forecasting. Finding the optimum parameter is very important, since the prediction performance might be increased in further studies for the various parameter differences [6].

Prediction of travel time can be done using k-nearest neighbor method in association of combined ATC system and VDS data [7]. Though ANN has an excellent previous history of prediction fields, there are more effective and practical technique which reduces the training time as ANN model usually takes long learning time [8]. Finding a historical traffic condition and compare them to the current condition is necessary for KNN. The main idea is to find historical traffic data which is like the current data. Their proposed k-NN model is combined of two pattern-matching steps. At first it matches the current data that can be obtained with historical travel time data. The historical data identified as like current data in the first step which executed by the second pattern matching [7]. The observed prediction shows an effect of weekdays and weekend data while comparing the prediction with other method. Moreover, unlike ANN and KF methods KNN does not need big training program so the model is easily transferable.

The Survey of queues in three selected banks in Ogun State, Nigeria [9] is first collected from and then tabulated in multiple tables based on each bank. The dataset was then used to estimate the average waiting time and queue length for each bank. The dataset was recorded using a stopwatch and a recorder and data on total number of customers served, service time of each customer and waiting time of each customer. This dataset was used to make prediction models to take decisions on how to increase servers, manage effectively queue length and service time. At the end of the research, the Analysis of variance (ANOVA) results of the dataset shows a significance of 0.05 of total time spent by the customers in three banks.

III. DATASET

The dataset used in this paper was obtained [9] and was retrieved from three banks from Ogun State, Nigeria. It comprises of service time and waiting time of each individuals which was obtained using a stopwatch. The specific dataset was selected because, queuing systems in sub-Saharan countries faces a lot of difficulties to maintain just like South Asian countries like Bangladesh, India, Pakistan and Sri Lanka. All these developing countries heavily depend on queues. As a result, due to huge

population, congestion occurs an efficient maintenance of queues is required.

The dataset was at first normalized into numerical values, excluding all categorical values and nominal values. At first, the weekdays were defined as numerical values between 1-7, where Sunday is 1 and Saturday is 7. The banks are given numerical values, such as the first bank is assigned 1 and the third bank was given 3. In the dataset, the service time and waiting time is represented in minutes as a result all the data were numerical beforehand. The main features considered are weekday, number of customers and service time in order to find the waiting time of customers.

Weeks	Bank	Weekday	Waiting_Time	Serving_time	Number_customers
1	1	2	12	26	880
1	1	3	5	19	720
1	1	4	6	8	1020
1	1	5	11	20	802
1	1	6	17	15	522
1	1	7	20	13	989
1	1	3	22	18	684
2	1	4	24	19	584
2	1	5	23	9	1021

Fig. 1. Dataset snapshot

The dataset being non-linear in nature was immediately chosen as a good resource for our research. Here, the non-linearity of the dataset required Algorithms that support non-linear dataset efficiently. As a result, to move forward in this research, the main algorithms that are chosen are, Support Vector Regression, K-Means Clustering [11] and KNN Classifier.

IV. PROPOSED METHOD

For addressing the problem and for obtaining accurate prediction from specific queue data, a very robust and less error prone prediction model is needed. For this very sole purpose, to identify and to predict a specific result, most suitable machine learning tool is needed to be used just like Support Vector Regression. Machine learning allows identification of data pattern and different libraries already available in different programming languages such as Python, allows easy implementation of a prediction model. The complete model is provided below:

Obtaining numeric queue data from a real-life queue, containing some real-life queue time metrics such as total served customers, waiting time, service time and weekday. This parameter is used for prediction. The queue data is obtained from real life scenarios from service counters and queue participants. Each participant with their individual work in the queue requires separate service time. As a result, prediction of service time is required beforehand. At first, a test implemented queue data is obtained from several real-life bank queues containing all the fields available Weekday, Queue Service Time, Queue Waiting Time, Total customer served for each individual queue. Individuals in the queue are considered in the dataset where all the related data are provided which will be required for future prediction. The very first field represents the specific day of the week, for which the record is documented. The next field, Queue Service time provides the time required for the individual to obtain the required service from the bank. The Queue

waiting time is the time waited by the customer in the queue and lastly total customer served gives the total number of customers already served in the queue. All the related fields in the dataset are in minutes as per calculated from the start of their participation in the queue. Both the service time and waiting time varies for each record based on their main objective in the queue.

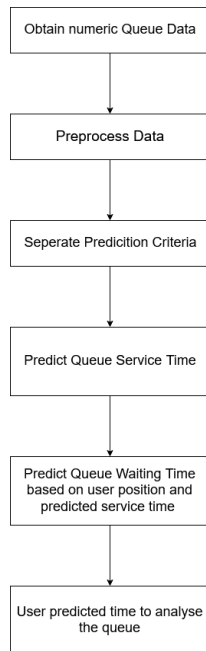


Fig. 2. Proposed Method

The data is pre-processed. Any irrelevant data, missing values as well as duplicate rows are identified and changed before any algorithm is applied. It provides the foundation of proper prediction and provides a certain prediction model for our prediction problem at hand. Separate prediction criteria, such as which column to consider during prediction is identified. Here at first, the data is split into test and train data, where the service time and position are taken into consideration. The testing data is used to find the accuracy and the training data frame is fed into a Support Vector Regression prediction model. The predicted service time is thus obtained.

The predicted service time is then used to predict the waiting time, and again the data frame is fed to Support Vector Regression prediction model. Here, at last for a position the predicted waiting time is obtained.

V. EXPERIMENTAL RESULTS AND DISCUSSIONS

The first experiment consisted of finding the most suitable machine learning tool for prediction. As a result, the dataset was evaluated with multiple classifiers. Most notably K-nearest Neighbors, k-Means clustering, Linear regression and Support vector regression. Here, the dataset was mainly evaluated in order to find the best suited form of solution.

For implementation the dataset was tested with a classifier K-Nearest Neighbors, two regression tools Support Vector Regression and Linear Regression and a clustering tool which is K-means clustering. For each classifier, the

accuracy score, mean squared error and variance score was obtained. Where it was found, the mean squared error of both Support Vector Regression and Linear Regression shows the most promising result when a prediction of waiting time with respect to Service Time was found.

K-Nearest Neighbors classifier was implemented with a variety of machine learning Python packages such as pandas, NumPy, pyplot from matplotlib and KNeighborsClassifier and preprocessing packages from scikit learn. At first the dataset in csv format is loaded using the pandas read_csv function. An input for the position, weekday and total customer served is taken from the user where the user in the queue inputs his position as n, weekday as w and total number of customers as c. After obtaining his/her input position, a numerical value that denotes his/her current position in the queue, a prediction for service time for a certain queue is obtained from the dataset, by classifying the dataset with respect to position, weekday and number of served customers. After the service time is obtained, the obtained predicted service time is then fed into another K-nearest neighbor classifier, for 7 nearest neighbors, and at last the value of predicted waiting time based on a predicted service time is obtained. The predicted waiting time obtained gives us our outcome, which is the probable waiting time for a person in a queue.

For Support Vector Regression (SVR) all the above procedures are followed, but here python packages SVR from Scikit Learn(sklearn) that stands for support vector regression is used. Here a linear SVR is implemented so that for infinite possibilities (as the outcomes are numbers), all the predicted outcomes can be obtained using a linear relation that makes the solution faster and efficient. SVR being a more complex yet, less error prone was our choice to implement. For linear regression, similar to the SVR, the python packages linear_model from sklearn was used. Here, the regression model is itself linear in nature and thus provides a simple regression model for prediction. For clustering the algorithm K-Means Clustering was chosen. Here the special package of scikit learn sklearn.cluster is used where K means is imported. For K means [11], the cluster size is set to 10 in order to obtain a higher scope of clustering and very approximate value to predict. Even though the prediction might provide a very general value for a specific cluster, but the algorithm is still implemented. The dataset for each individual Algorithm was then evaluated and the accuracy score, mean squared error and also variance of each classifier is obtained. The obtained results are given below:

TABLE I. PERFORMANCE EVALUATION OF CLASSIFIERS

Classifier	Accuracy %	Mean Squared Error	Variance Score
K-Means	25	202	2.4
SVR	86	12.94	0.43
Linear Regression	67	32.28	0.6
KNN	33	79.53	1.65

From the above table we can observe in case of k nearest Neighbor, an accuracy of 33% is obtained with mean squared

error of 202 and variance score of 2.4. For SVR, the accuracy rate was quite impressive, which is 86% with low mean squared error of 12.94 and variance of 0.43. For the linear regression model an accuracy of 67% with mean squared error of 12.94 and variance score of 0.6 was obtained. Finally, for KNN, an accuracy of 25% with mean square error of 202 and a variance score of 2.4. From the above table, it was found SVR having a high accuracy of 86% and a variance score of 0.43 which is a good measure for highly sensitive data. Making the prediction model accurate as well as robust. Here even the value of mean squared error is only 12.94. As a result, it was verified Support Vector Regression as an ideal tool for queue waiting time prediction. The first experiment consisted of feature selection and dataset modification. For finding the pattern of service time and waiting time with respect to a queue position. For this, another dataset was generated for weekday=3(Tuesday) and total number of customers=100 served as constant. This helps in generating a dataset composed of results found for all the classifiers.



Fig. 3. Scatter plot for service time(x-axis) with respect to waiting time (y-axis) from dataset

Scatter plot is first used to evaluate the relation between service time and waiting time in queue. Here from this data visualization it was found, both the attributes are completely independent from each other. As a result, a prediction for service time beforehand is required before predicting the waiting time. Here, the data has no correlation as a result for prediction of one, the value of the other needs to be found beforehand. The prediction model for predicting the service time with respect to time is then measured using three different algorithms, K Means Clustering, k-Nearest Neighbor and Support Vector Regression. At first the prediction model of SVR is plotted using a bar plot where the position is kept in x-axis and predicted waiting time in y-axis. The figure provided a linear prediction model where with increase of position the value of predicted waiting time also increases. This model provided a very solid and accurate prediction model. The relation between position and waiting time proves directly proportional and thus proves a great dataset for regression and predicting Service waiting time.

The next representation is for K-Means and here we obtain a bar plot for predicted waiting time with respect to position. But here we obtain a bar plot that shows irregular variations of prediction which looked very unlikely for a

prediction model for the problem at hand. For example, the predicted waiting time for position 4 is 7 where for 7 is also 7. As a result, the prediction obtained by K-Means clustering proves repetitive and less accurate as per evaluation as well as visualization. In case of KNN, the bar plot obtained shows irregularity. Here the result obtained which is the waiting time is comparatively is very low. Here the result obtained is so irregular such that for both position 1 and 10 the waiting time is 0. And also, for position 8 the waiting time is incredibly high which is 63. As a result, this model also does not provide an ideal prediction model for our problem.

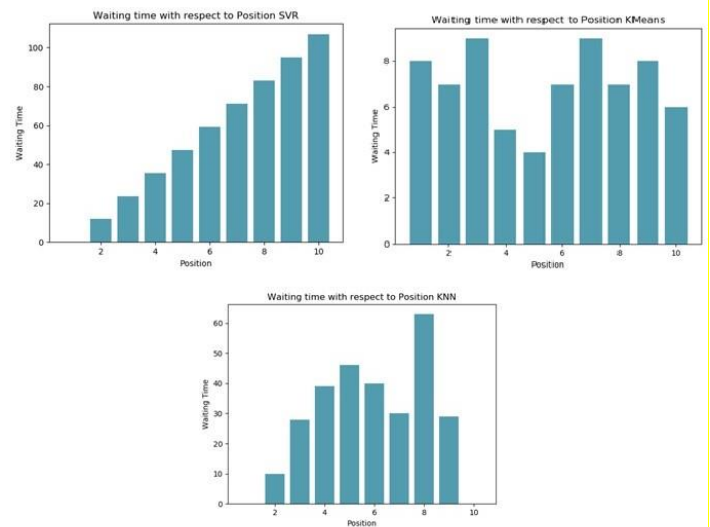


Fig. 4. Bar plot of Predicted waiting time for first 10 positions using K-Means Clustering, KNN Classifier and SVR

Comparing all the above algorithms, it was found SVR proved the most accurate and optimum solution to the problem at hand.

VI. CONCLUSION

Support Vector Regression has previously demonstrated its success in time-series analysis and for predicting various important data based on preprocessed data. Here in this paper, the algorithm has also shown promising results. Here a fundamental model for queue waiting time prediction depended on predicted service time from training dataset has been established. Different machine learning algorithms have been implemented using different machine learning approaches on a specific dataset from which SVR has shown for the best results in comparison with other approaches. The result shows the outstanding performance of SVR for waiting time of a queue. Yet there is a big scope for queue analysis to be explored in near future to predict using other attributes also. There is also a scope to study more and find a better solution to queuing problems that provides a more accurate result above 86% unlike Support Vector Regression in this paper. Moreover, the results obtained from this research can be used to further enrich similar systems and solutions to this banking queuing related problems. This research can be made more effective and efficient by using more varied feature set and selecting more appropriate predicting features. In order to improve the overall approach, new

improvements related to the dataset can be obtained. More new datasets can be taken to evaluate and compare the approaches and evaluate the model proposed in this paper. The model can be tested in a more real- life scenario, where a real time dataset can be used for prediction in real life. All the above is a part of our future goal for the current research. Due to the success of SVR in improving the accuracy score and minimize Mean Squared Error, it can be concluded the algorithm is both accurate and efficient at the same time. The model proposed can be further extended and be used in other sectors besides banking, given a detailed and accurate dataset is given. The research aims to provide a cornerstone to the research of prediction that now can be stated based on the results of this research, in hope that the study of Prediction of patterns and numerical values can be further improved soon.

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